



HTML & CSS

Lecture 14





TABLE OF CONTENT

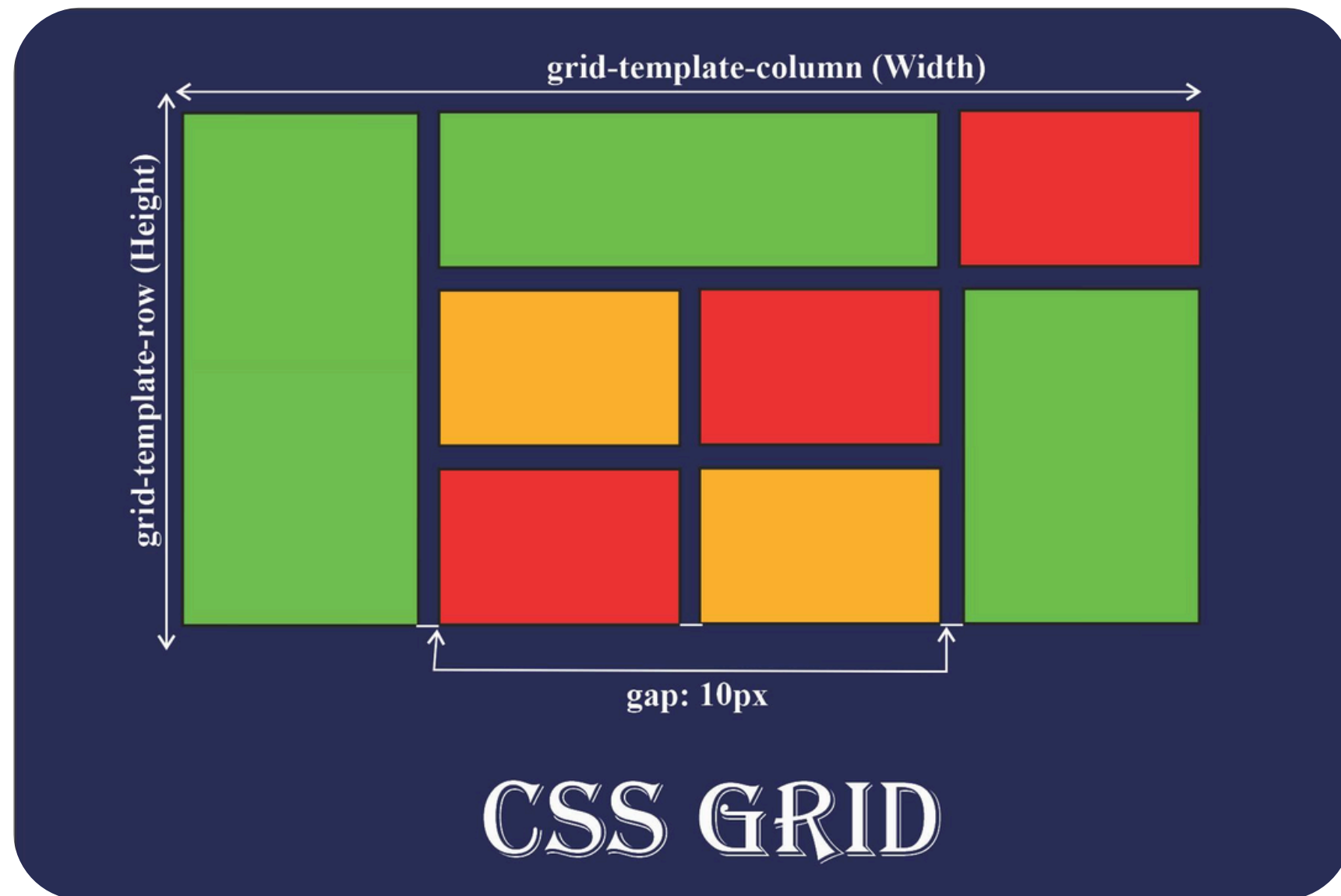
- ▶ Display Grid
- ▶ Grid container
- ▶ Grid items





Display Grid

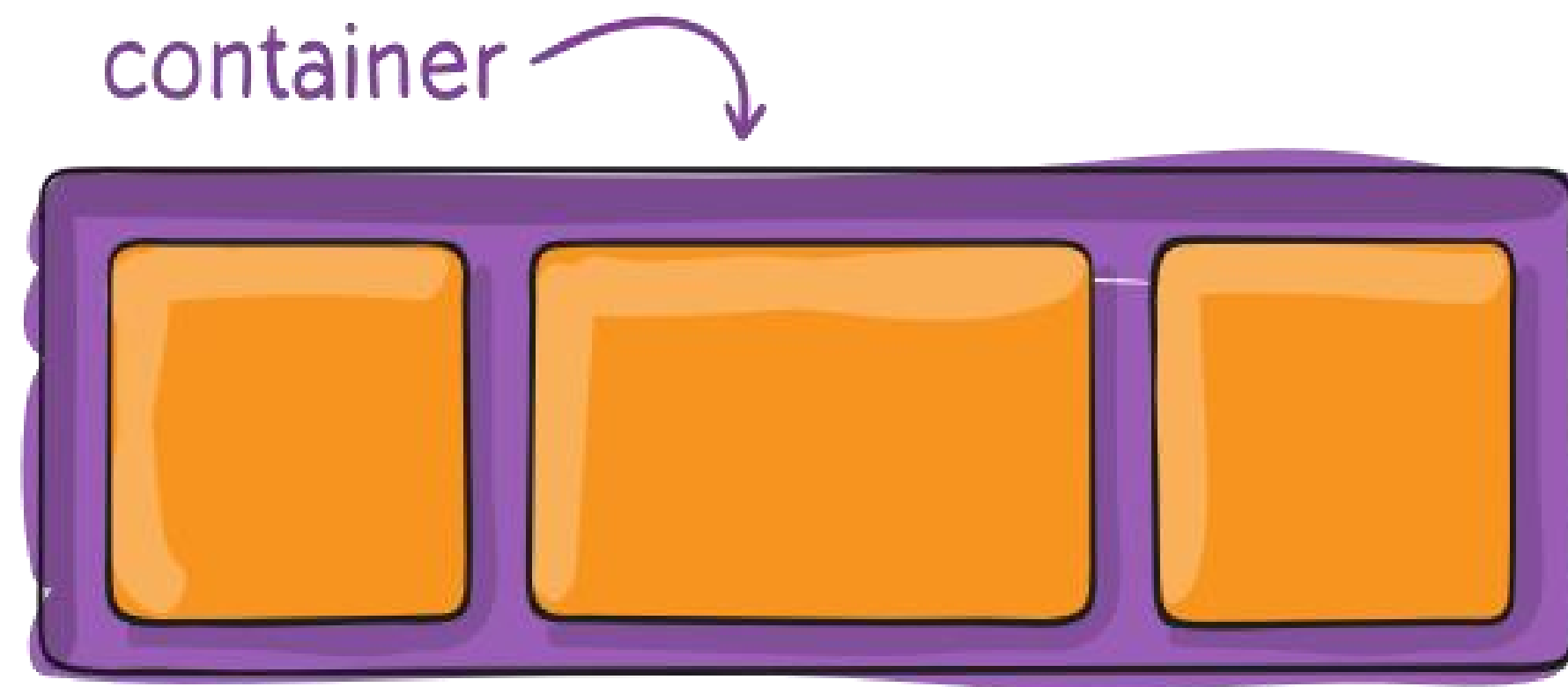
Display Grid



Grid is the very first CSS module created specifically to solve the layout problems.

CSS Grid Layout is a two-dimensional grid-based layout system that, compared to any web layout system of the past, completely changes the way we design user interfaces.

Properties for the Parent (Grid container)



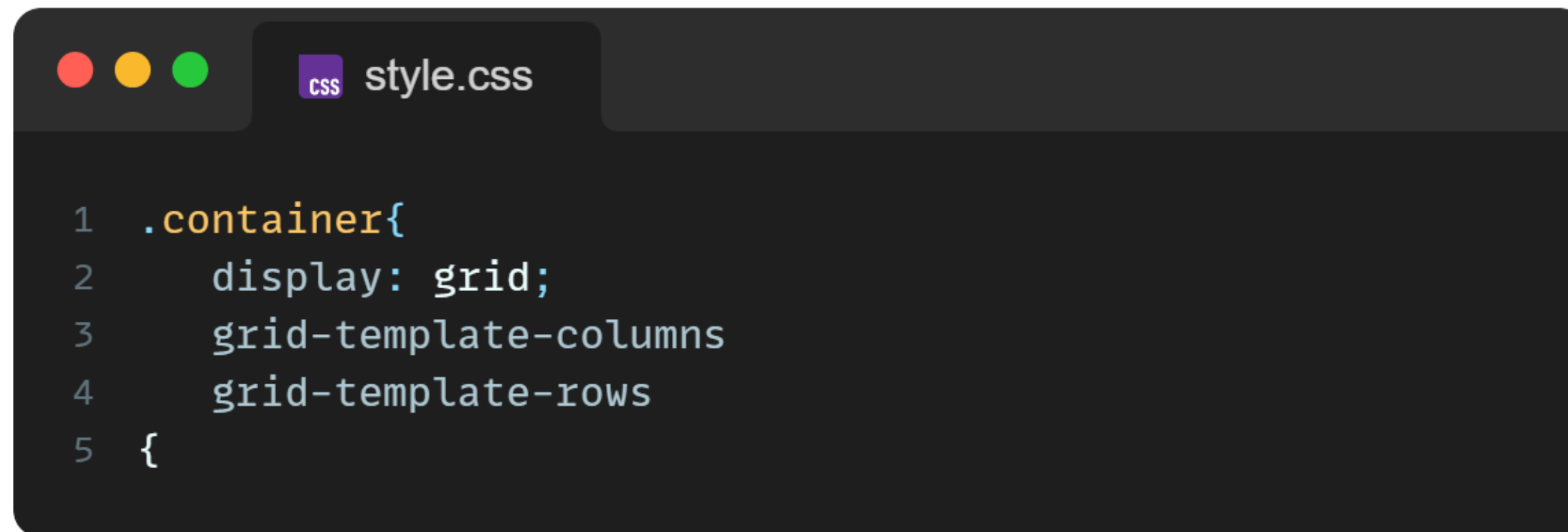
Display

A code editor window with a dark background. The title bar shows three colored circles (red, yellow, green) and a tab labeled 'style.css' with a 'css' icon. The code is as follows:

```
1 .container{  
2     display: grid;  
3 }
```

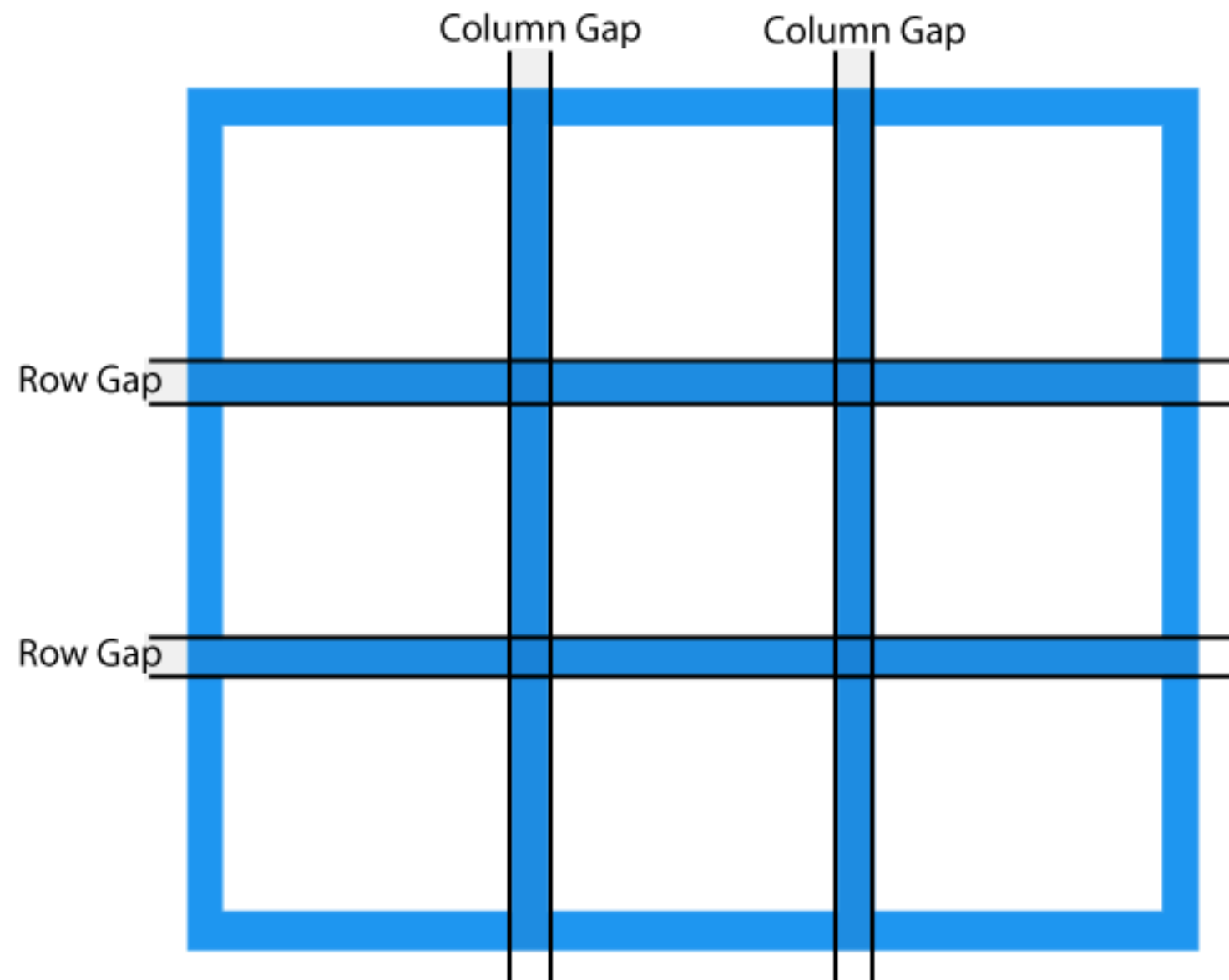
Defines the element as a grid container and establish a new grid formatting context for its content

- **grid-template** - columns - creates columns in our container
- **grid-template** - **rows** - creates rows in our container

A code editor window with a dark background. The title bar shows three colored circles (red, yellow, green) and a tab labeled 'style.css' with a CSS icon. The code is as follows:

```
1  .container{
2      display: grid;
3      grid-template-columns
4      grid-template-rows
5  {
```

Grid gaps (gutters)



```
1 .container{
2   gap: column-gap row-gap;
3 }
```

The **column gap** and **row gap** creates a gutters between elements. Shorthand for this is **only gap**

Justify-items

```
.container {  
  justify-items: start;  
}
```



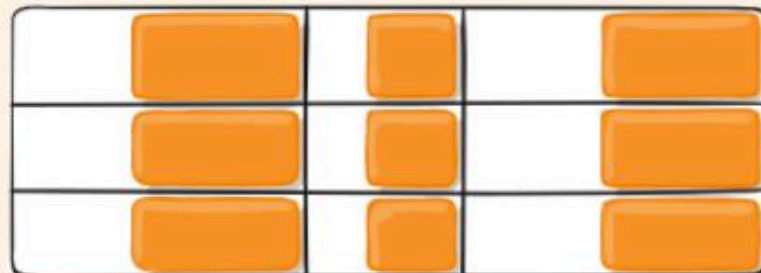
```
style.css  
  
1 .container{  
2   justify-items: start | center | end | stretch;  
3 }
```

```
.container {  
  justify-items: center;  
}
```



Justify items control elements in their own boxes with row direction

```
.container {  
  justify-items: end;  
}
```

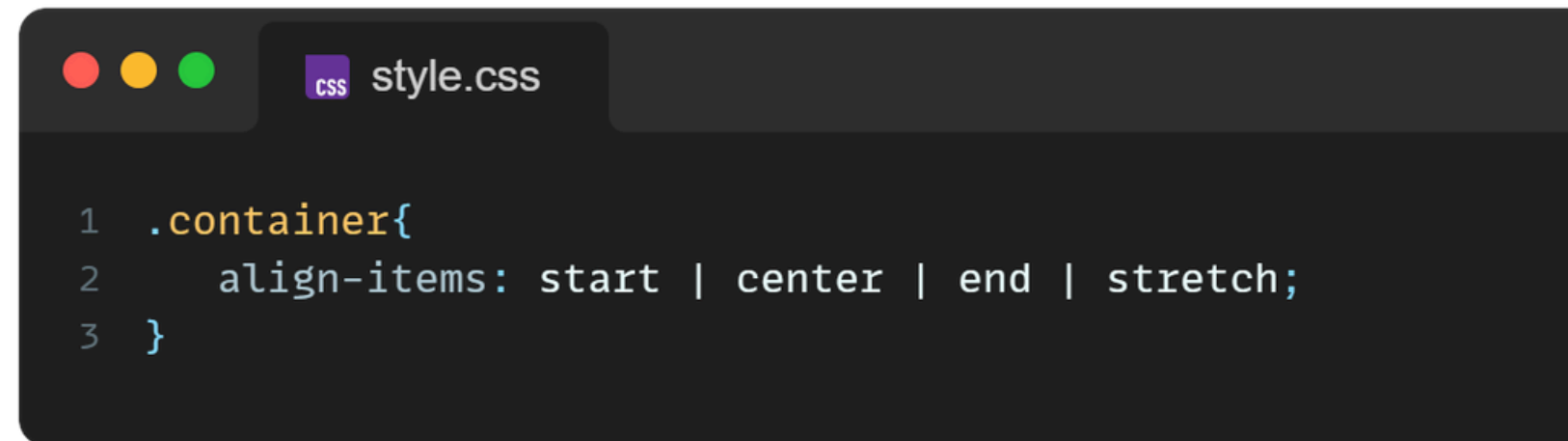
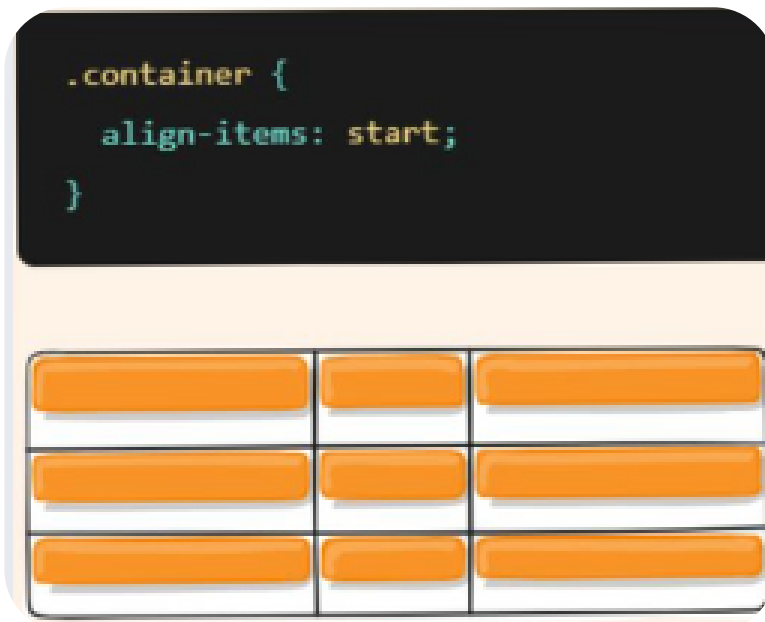


- **stretch** - default value
- **start** - justify elements to the start
- **center** - to the center of the box
- **end** - to the end

```
.container {  
  justify-items: stretch;  
}
```

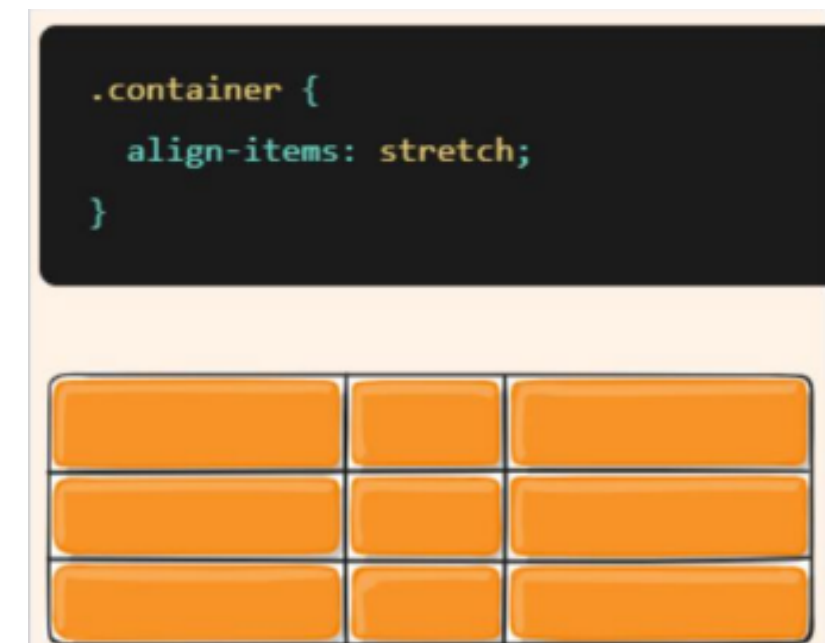
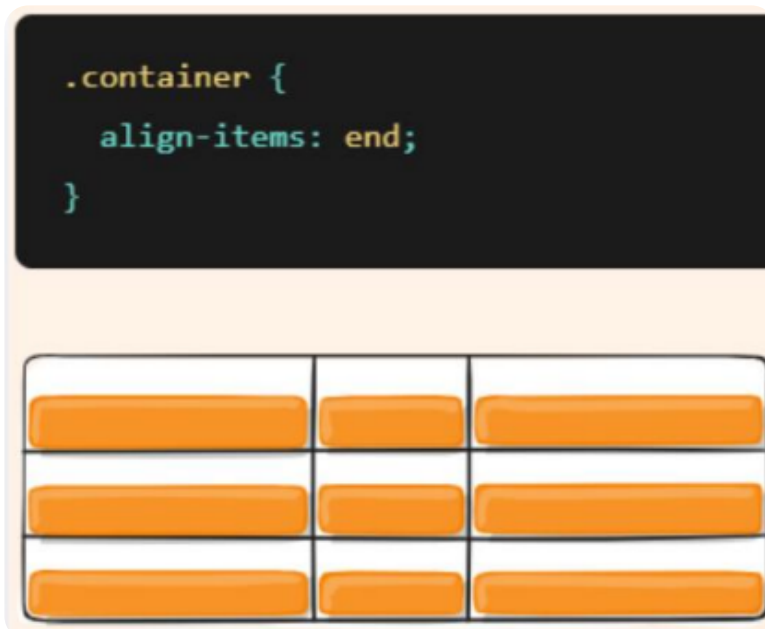


Align-items



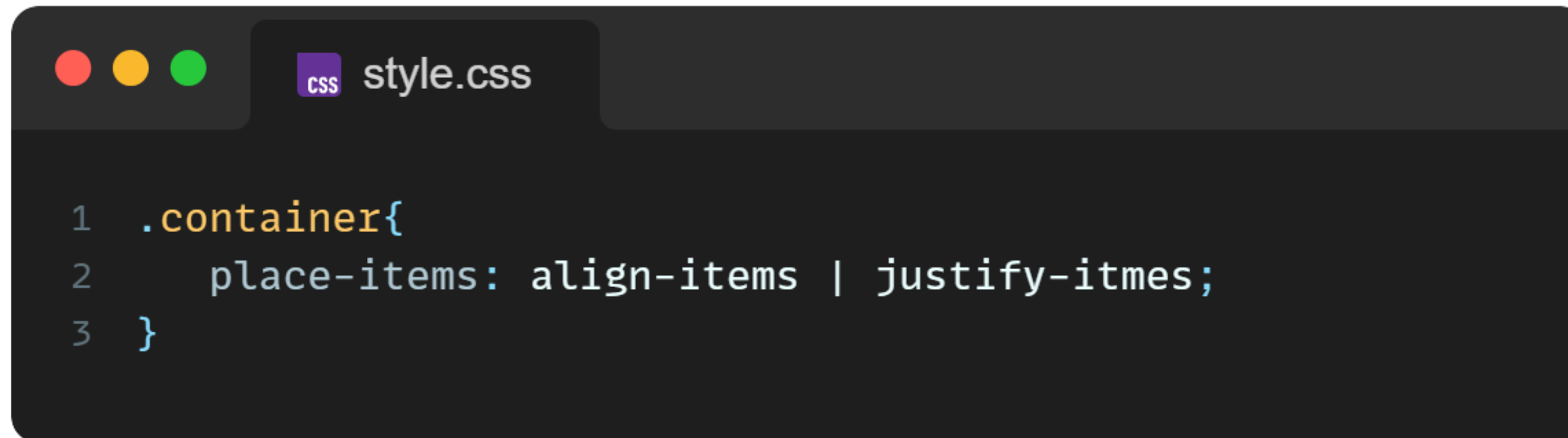
align items control elements in their own boxes with column direction

- **stretch** - default value
- **start** - justify elements to the start
- **center** - to the center of the box
- **baseline** - make line with texts
- **end** - to the end



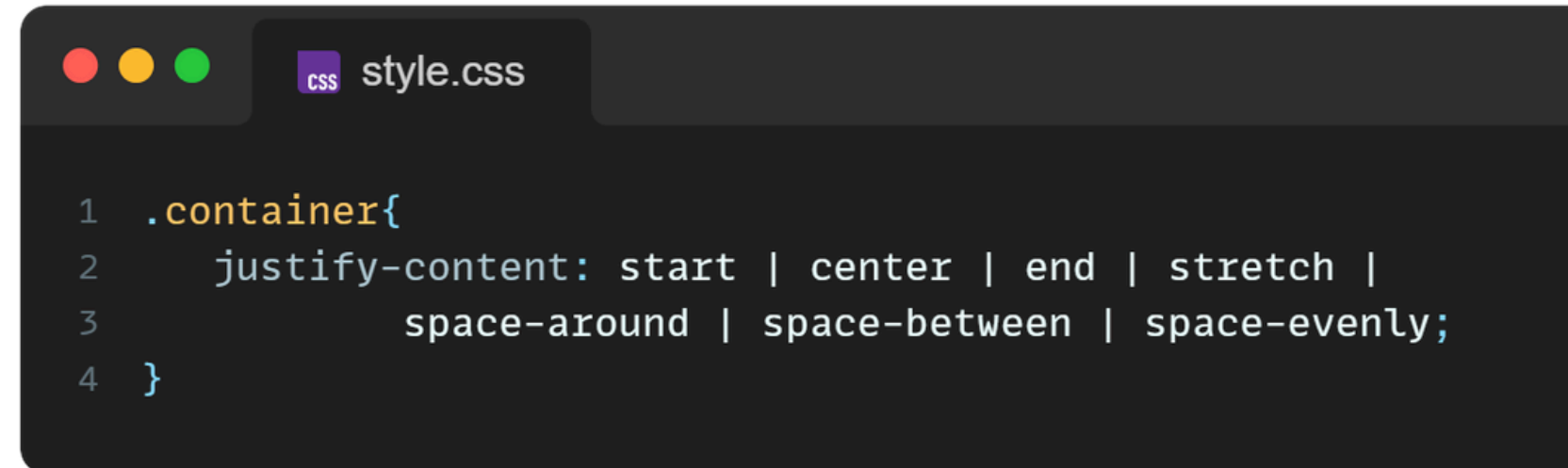
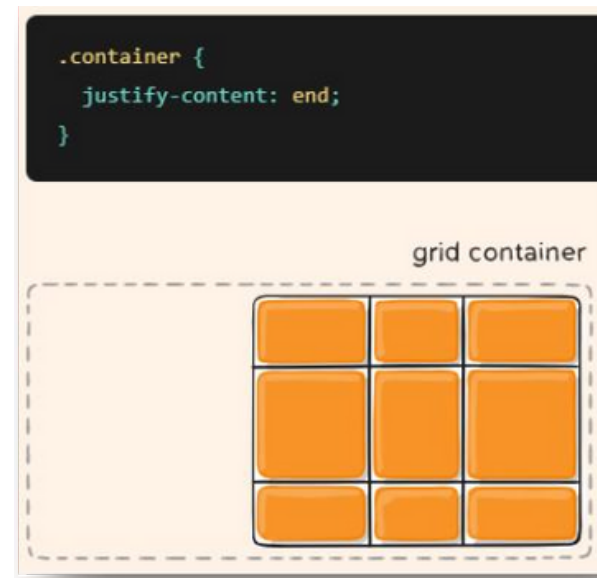
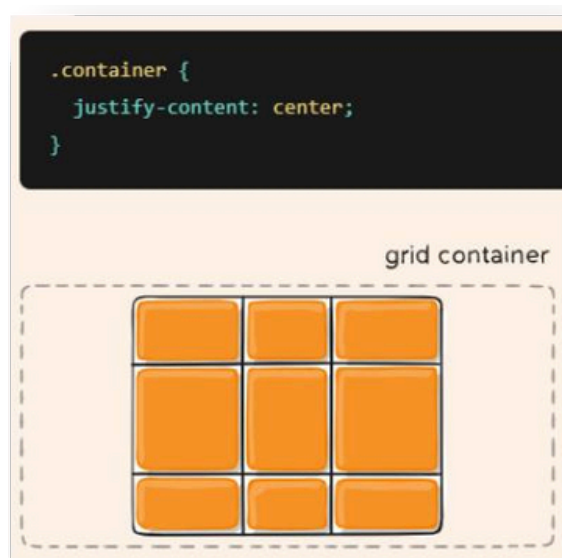
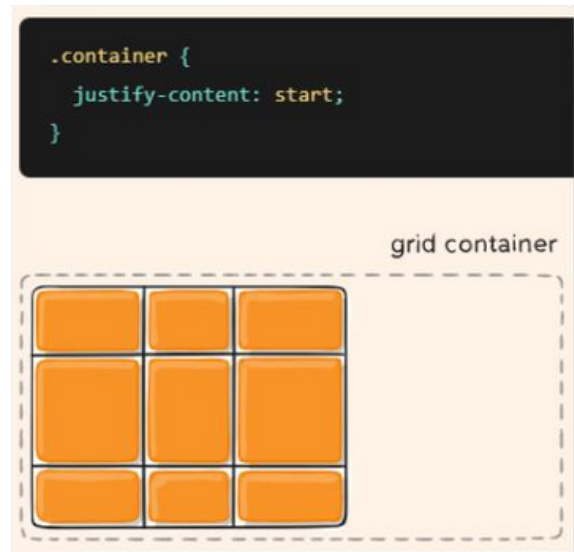
Place-items

The **place-items** property sets both the align-items and justify-items property at the same time

A code editor window with a dark theme. The title bar shows three colored circles (red, yellow, green) and a tab labeled 'style.css' with a purple 'css' icon. The code is as follows:

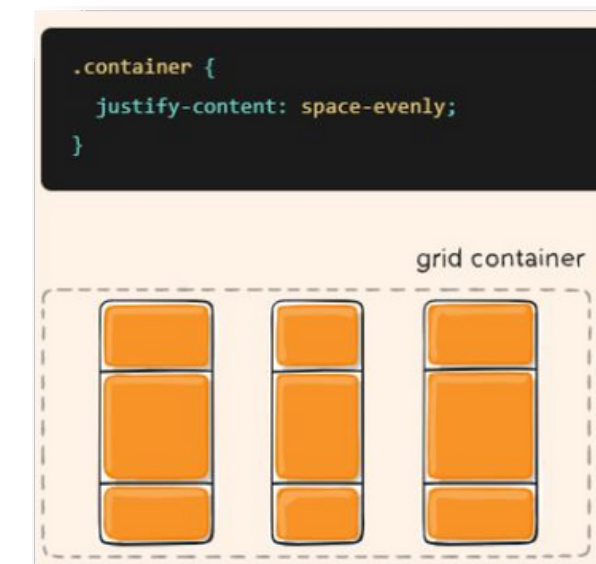
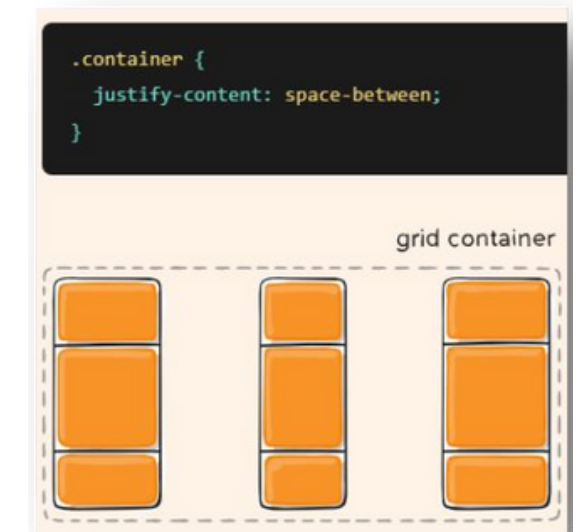
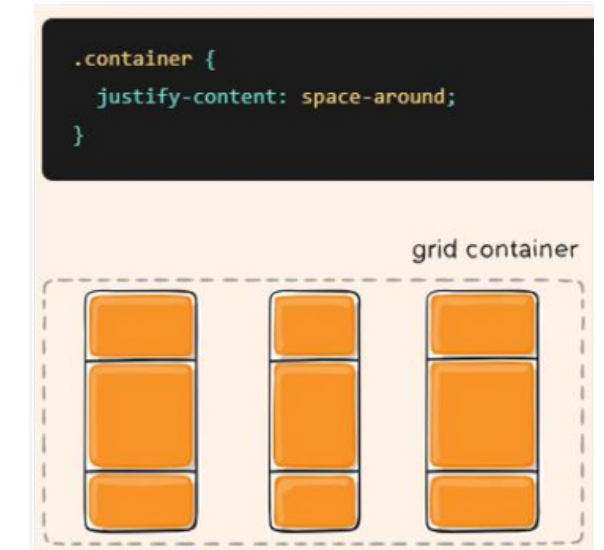
```
1  .container{
2      place-items: align-items | justify-items;
3  }
```

Justify-content



Justify content control columns in grid container

- **stretch** - default value
- **start** - justify elements to the start
- **center** - to the center of the box
- **end** - to the end
- **space-around** - space less between and sides
- **space-between** - spaces all in sides
- **space-evenly** - equal space in all side

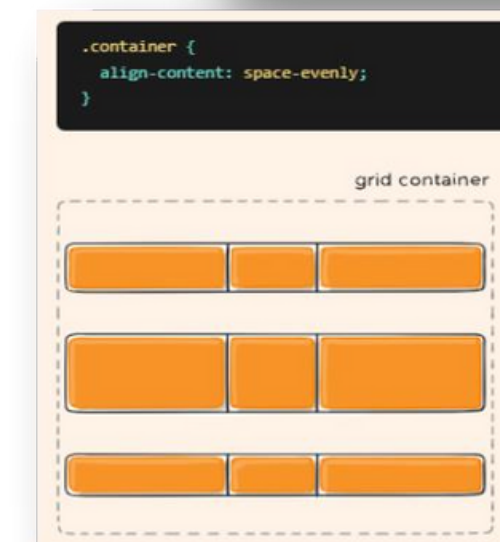
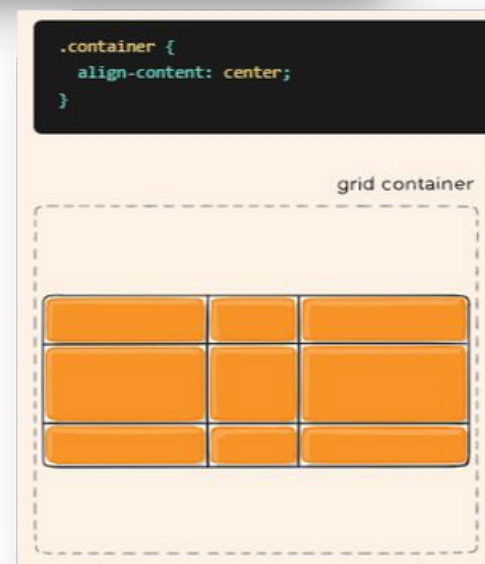
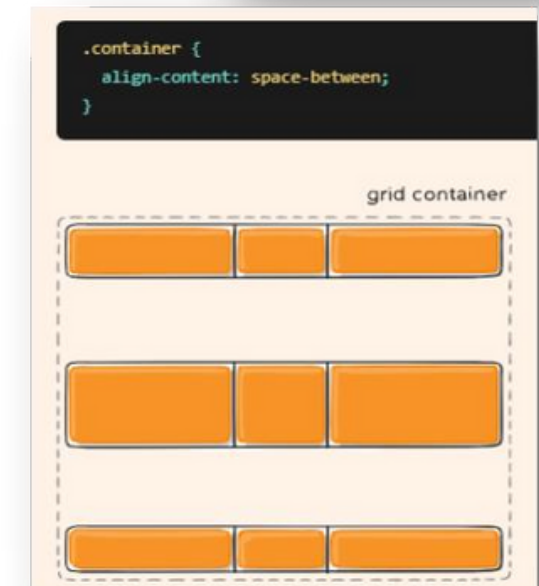
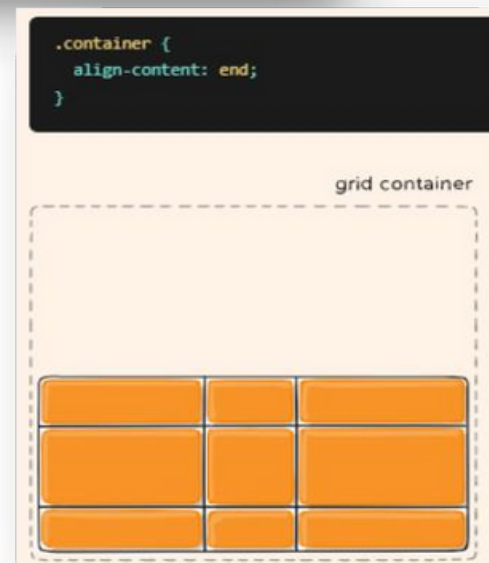
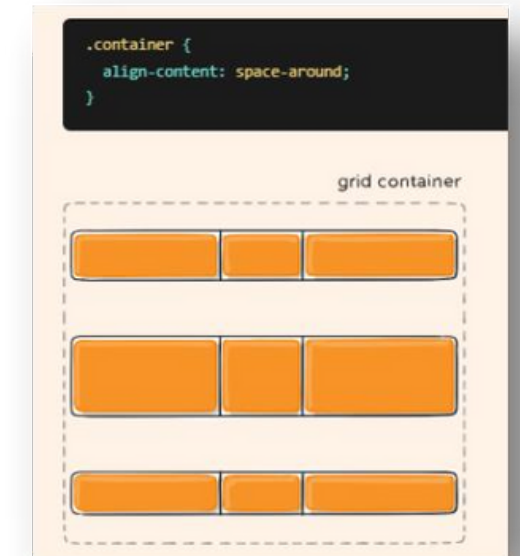
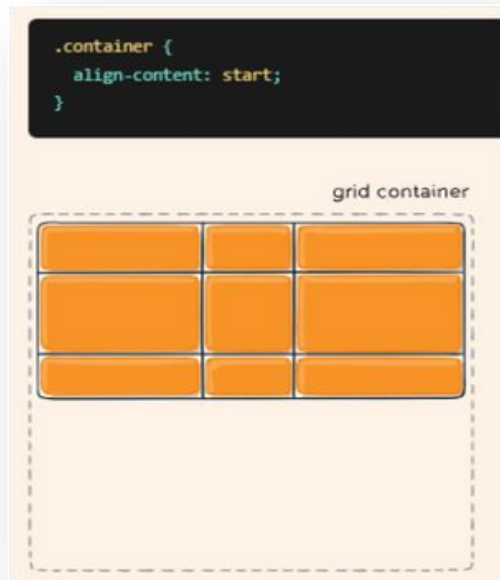


align-content

```
style.css
1 .container{
2   align-content: start | center | end | stretch |
3     space-around | space-between | space-evenly;
4 }
```

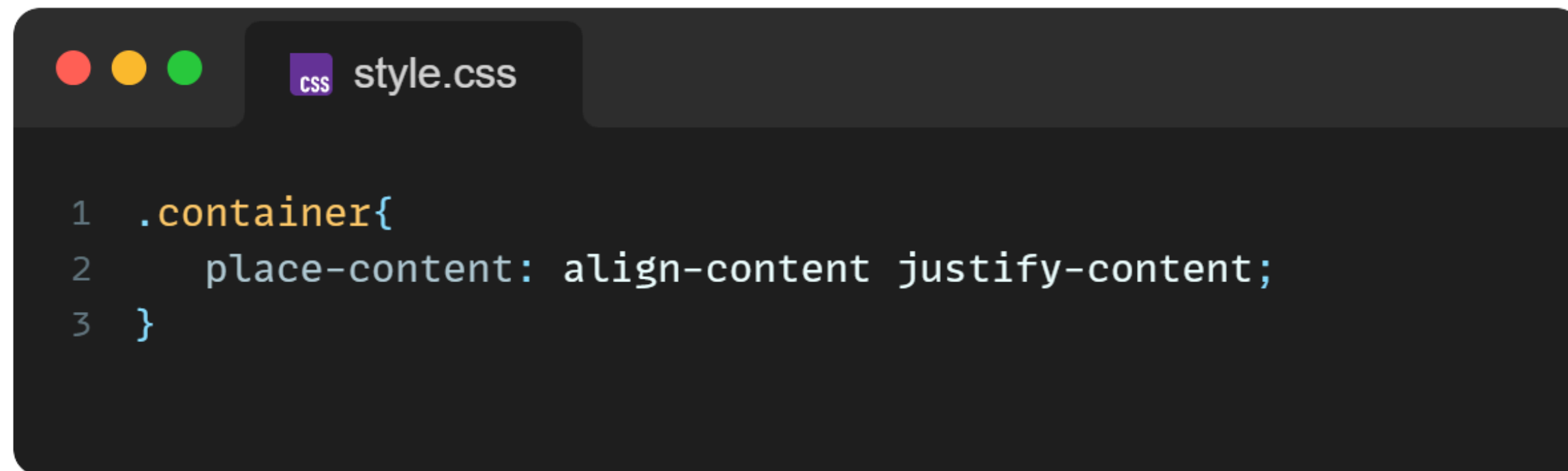
align content control rows in grid container

- **stretch** - default value
- **start** - justify elements to the start
- **center** - to the center of the box
- **end** - to the end
- **space-around** - space less between and sides
- **space-between** - spaces all in sides
- **space-evenly** - equal space in all side



Place-content

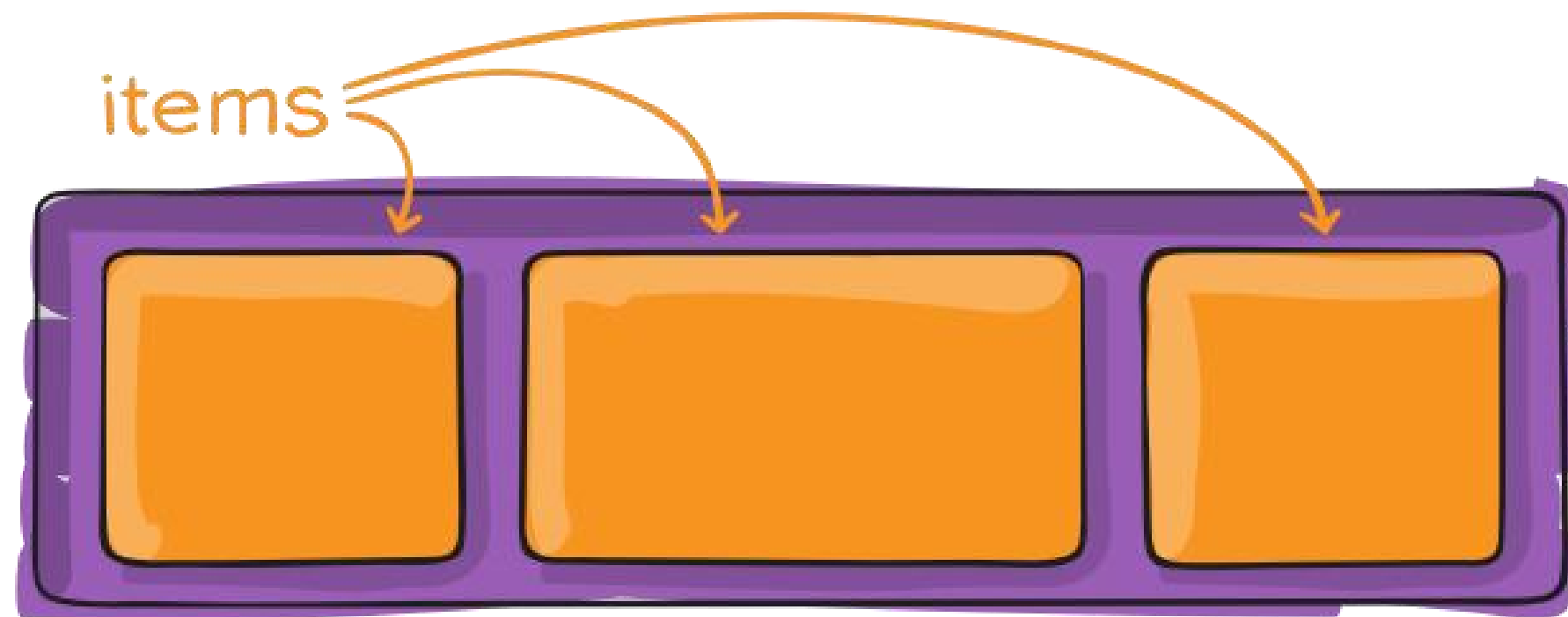
The **place-content** properties sets both the align-content and justify-content property at the same time

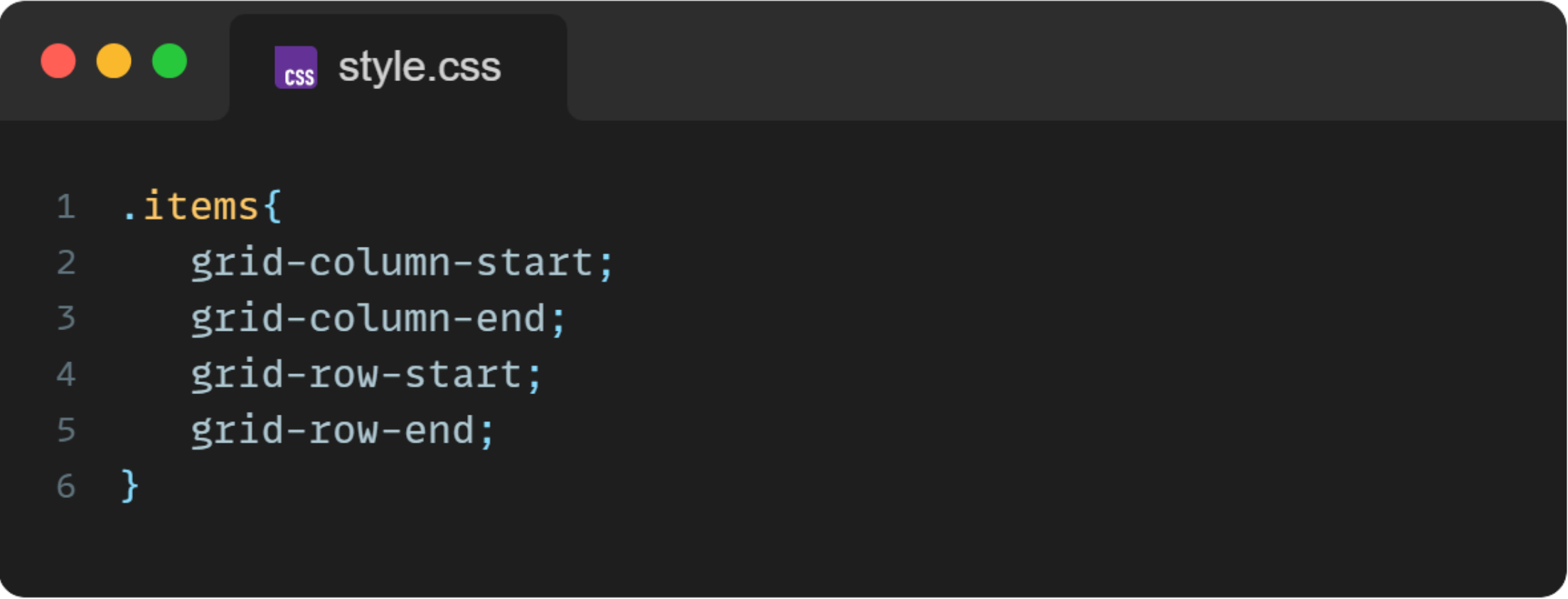
A code editor window with a dark background. The title bar shows three colored circles (red, yellow, green) and a tab labeled 'style.css' with a purple 'css' icon. The code is as follows:

```
1  .container{  
2      place-content: align-content justify-content;  
3  }
```

The code defines a CSS class `.container` and sets the `place-content` property to `align-content justify-content`.

Properties for the children (grid items)





```
1 .items{
2     grid-column-start;
3     grid-column-end;
4     grid-row-start;
5     grid-row-end;
6 }
```

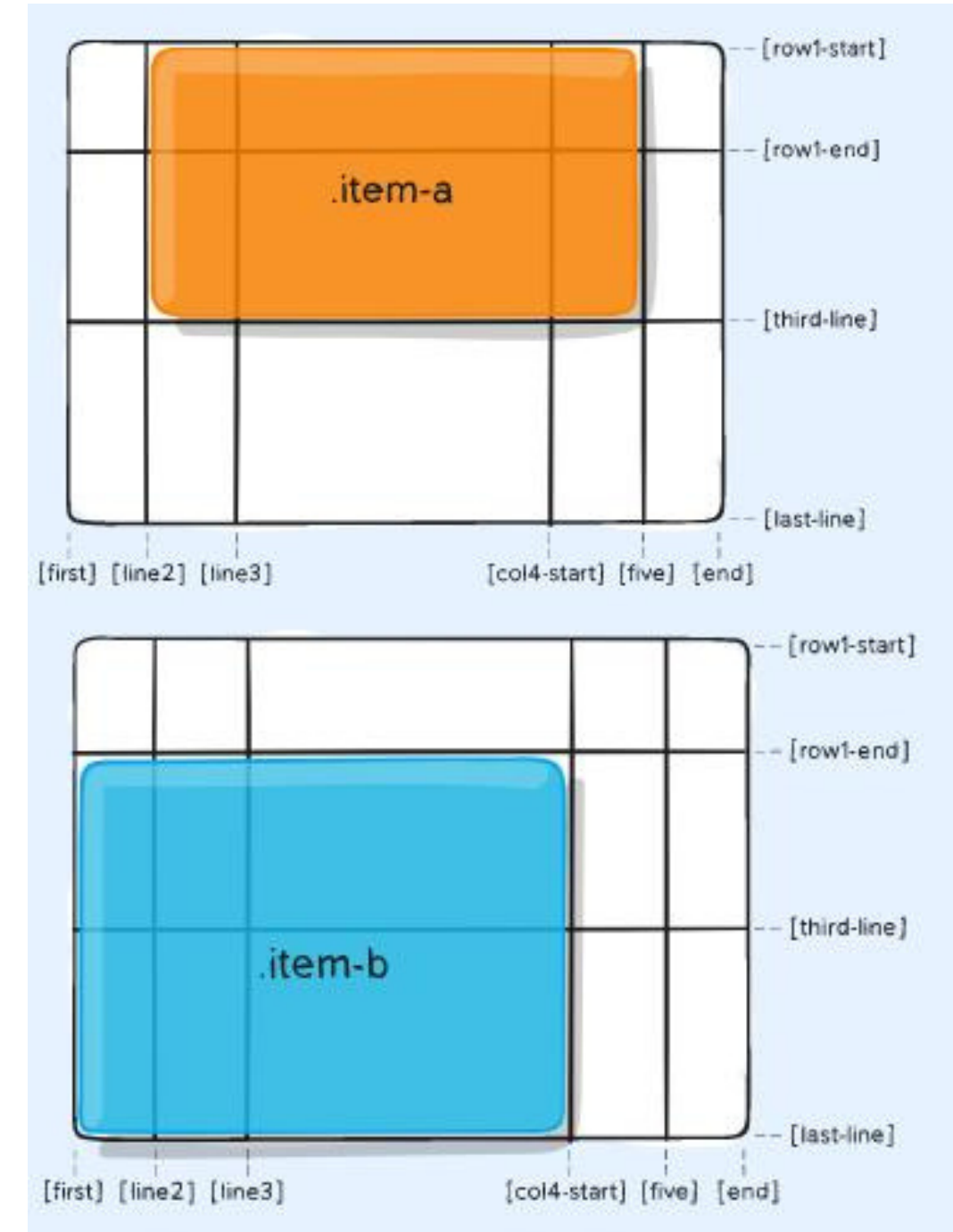
Defines the grid items location from which **column** should **start** and **end** and from which **row** it should **start** and **end**;

Code

```
style.css

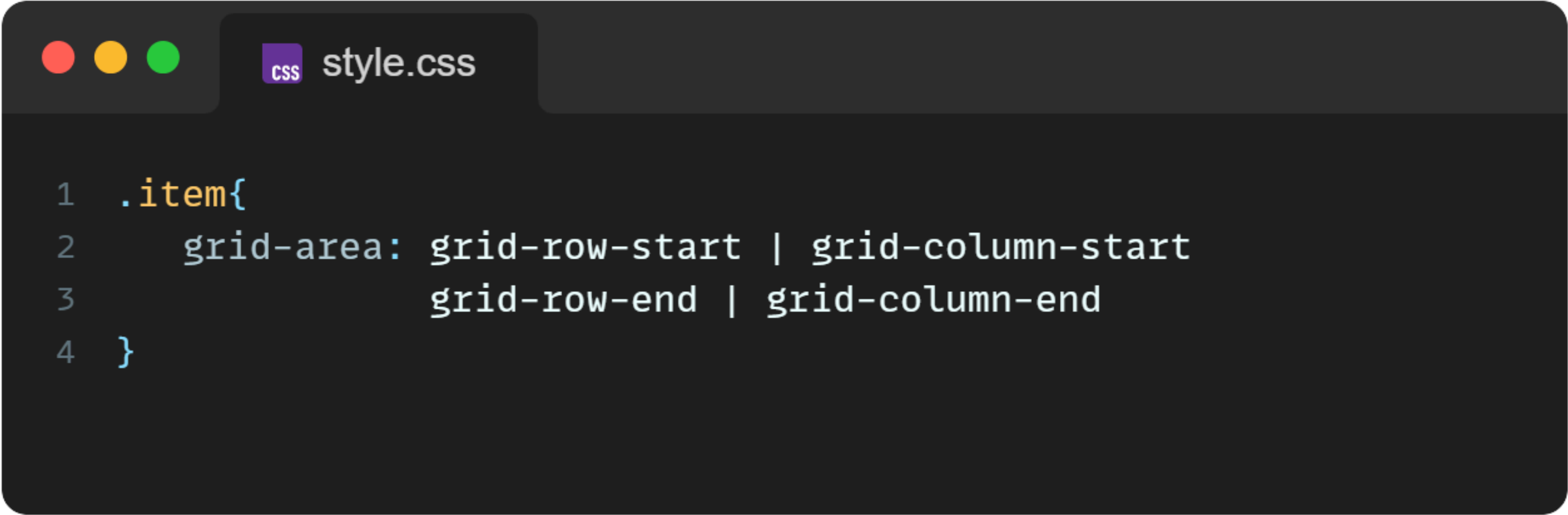
1  .item-a{
2      grid-column-start: 2;
3      grid-column-end: 5;
4      grid-row-start: 1;
5      grid-row-end: 3;
6  }
7  .item-b{
8      grid-column-start: 1;
9      grid-column-end: 4;
10     grid-row-start: 2;
11     grid-row-end: 4;
12 }
```

Result



Grid-area

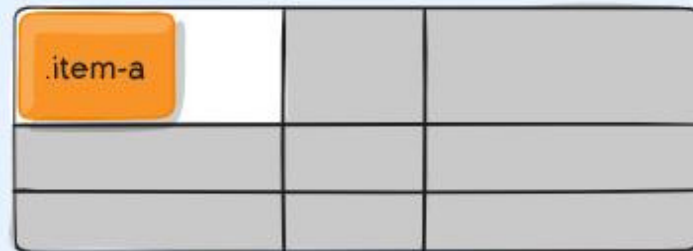
The grid-area property set all grid-column and grid-row properties;

A code editor window with a dark background. The title bar shows three colored circles (red, yellow, green) and a tab labeled 'style.css' with a 'css' icon. The code is as follows:

```
1  .item{  
2      grid-area: grid-row-start | grid-column-start  
3          grid-row-end | grid-column-end  
4  }
```

Justify-self

```
.item-a {  
  justify-self: start;  
}
```



```
style.css  
  
1 .item{  
2   justify-self: start | center | end | stretch;  
3 }
```

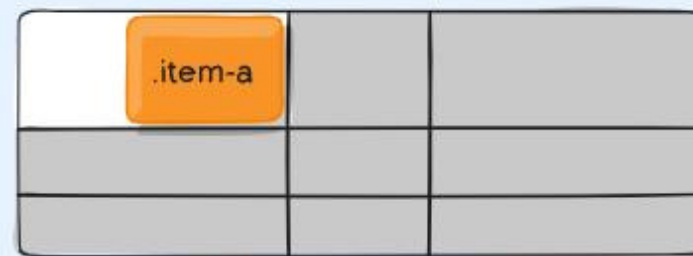
```
.item-a {  
  justify-self: center;  
}
```



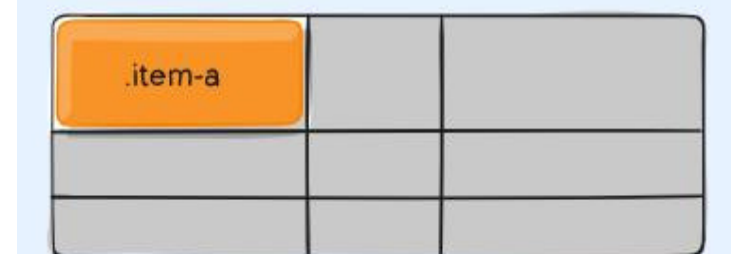
Justify self control one element in row direction;

- **stretch** - default value
- **start** - justify elements to the start
- **center** - to the center of the box
- **end** - to the end

```
.item-a {  
  justify-self: end;  
}
```



```
.item-a {  
  justify-self: stretch;  
}
```



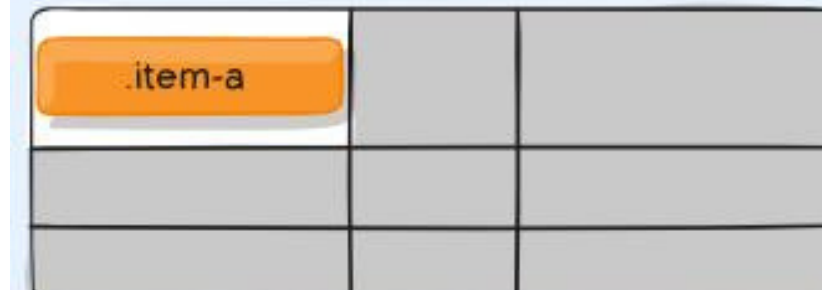
align-self

```
.item-a {  
  align-self: start;  
}
```



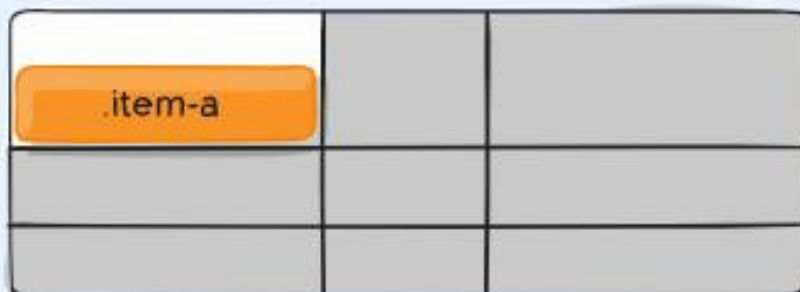
```
1 .item{  
2   align-self: start | center | end | stretch;  
3 }
```

```
.item-a {  
  align-self: center;  
}
```



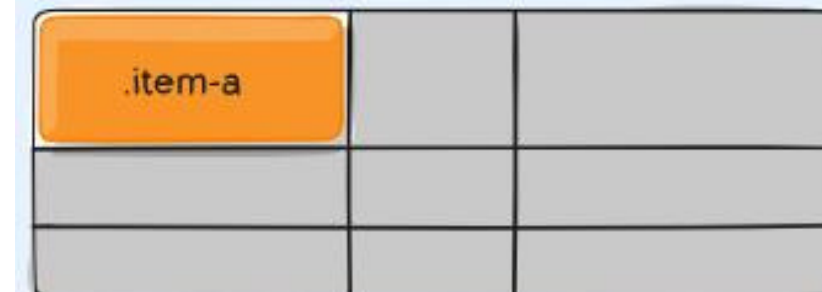
align self controll one element in column direction;

```
.item-a {  
  align-self: end;  
}
```



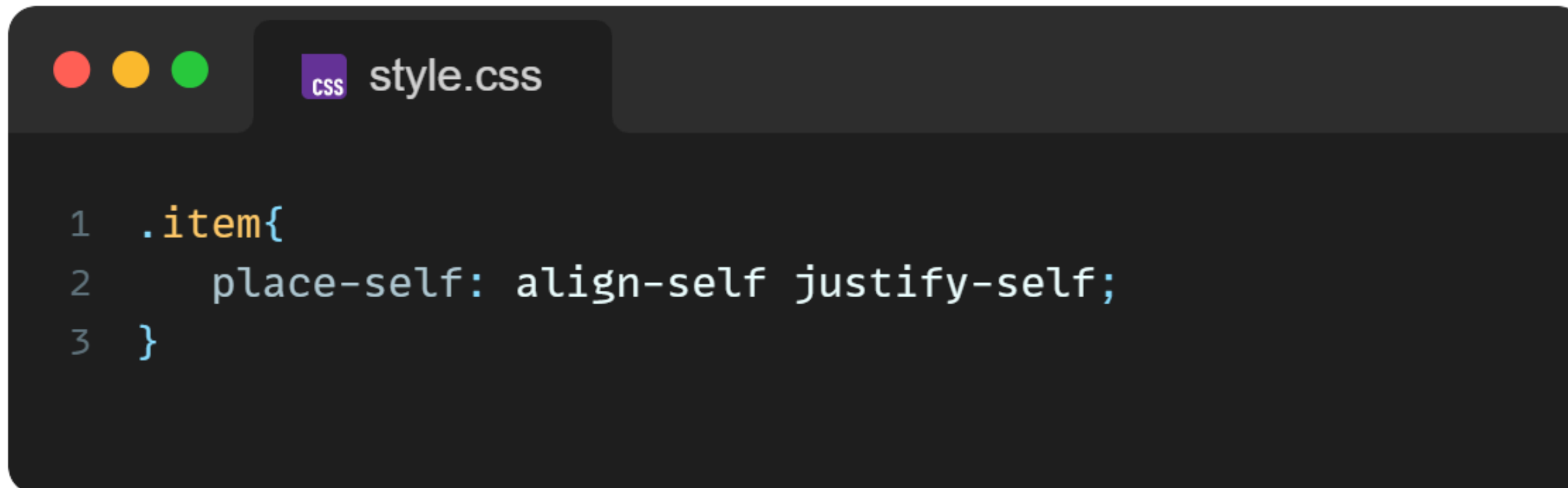
- **stretch** - default value
- **start** - justify elements to the start
- **center** - to the center of the box
- **end** - to the end

```
.item-a {  
  align-self: stretch;  
}
```



Place-self

The **place-self** properties sets both the align-self and justify-self property at the same time

A code editor window with a dark background. The title bar shows three colored circles (red, yellow, green) and a tab labeled 'style.css' with a 'CSS' icon. The code is as follows:

```
1  .item{  
2      place-self: align-self justify-self;  
3  }
```


THANK YOU!

“Start where you are. Use what you have. Do what you can.”

